

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457031.8705 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.048
Transit depth (Rp/Rs)^2	2.67 +/- 1.55 [%]
Orbital Inclination (inc)	85.96 +/- 2.86
Airmass coefficient 1 (a1)	13.12 +/- 9.66
Airmass coefficient 2 (a2)	-1.67 +/- 1.53
Scatter in the residuals of the lightcurve fit is	86.44 %
Best Comparison Star	None
Optimal Aperture	0.82
Optimal Annulus	4.89
Transit Duration (day)	0.088 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.048 vs 0.1368 (deviation 0.55σ)
Duration fit vs published	0.088 d vs 0.0758 d (deviation 0.94σ)
Mid-time O–C	-3.0 min
Depth SNR	3.4
Residual scatter	86.44 %

Light curve

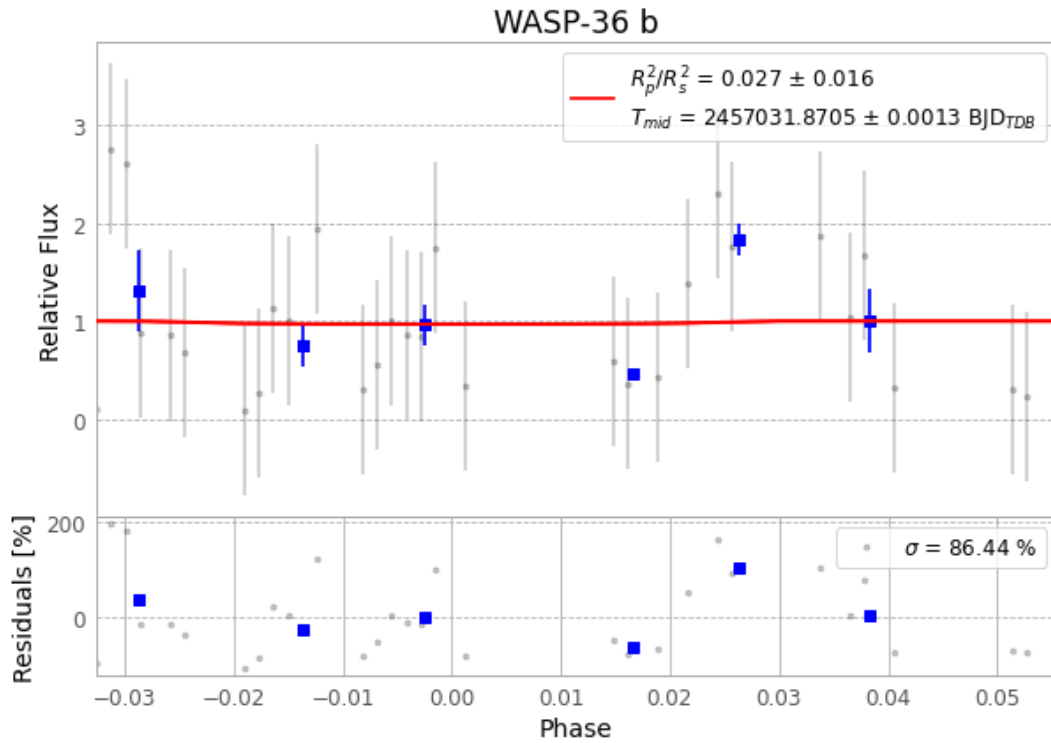


Figure 1 — FinalLightCurve_WASP-36 b_2015-01-09.png (SHA-256 e2ca9d1ee51b6d45...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [471, 477], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7604f352b3906637...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

68 FITS frames (45 MB), WASP-36150109073012.FITS → WASP-36150109105112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-150109.log (a76e2587871c...), FinalLightCurve_WASP-36 b_2015-01-09.png (e2ca9d1ee51b...), FinalLightCurve_WASP-36 b_2015-01-09.csv (95fc3ac8d1e6...)

Compute resources

Wall time 131.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:43Z.